

Central Veno-Arterial ECMO Physiology

We will now continue our exploration of VA ECMO physiology by considering the physiology of centrally cannulated VA ECMO. The proportion of central cannulation is largely facility dependent. While some programs mainly rely on peripherally placed ECMO cannulas, others have a significant percentage of central cannulation. Regardless, it is important to understand the physiologic subtleties unique to this configuration.

PRIMARY CENTRAL CANNULATION CONFIGURATION

As we learned in our configurations discussion in [Chapter 8](#), central cannulation involves cannulation directly to the heart and great vessels. The usual configuration is a dual-stage cannula implanted directly to the right atrium with a cannula implanted directly onto the aorta with an additional vent of the left ventricle either by a drain of the pulmonary veins or directly from the left ventricular apex ([Fig. 15.1](#)).

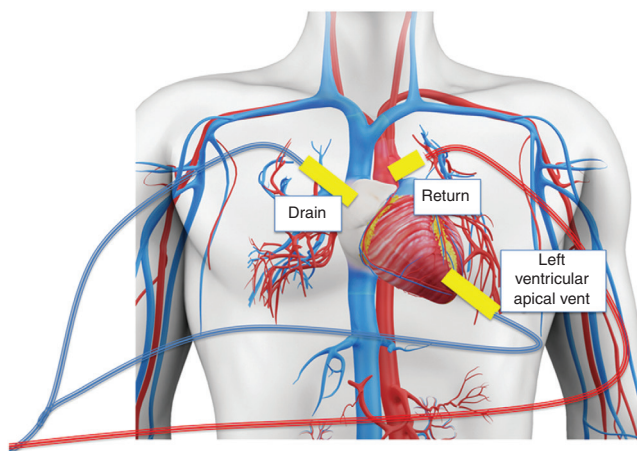


FIG. 15.1 Central cannulation configuration. (Modified from [SciePro/Shutterstock.com](#).)

ANTEGRADE FLOW AND PERFUSION OF CORONARY ARTERIES

The result is the ability to provide antegrade flow that bypasses both the heart and lungs. The physiologic advantages over retrograde support should be immediately apparent. Retrograde flow is no longer required, meaning that everything that is flowing from the extracorporeal membrane oxygenation (ECMO) circuit is what the body is receiving. The adverse effects of retrograde flow

that we discussed in [Chapter 14](#) (left ventricular afterload, competing flows, upper body hypoxia, etc.) are significantly mitigated in this configuration. Specifically, perfusion of the cerebral vessels is much more reliable, as the aortic cannula will be providing antegrade flow prior to the takeoff of the innominate and left common carotid artery ([Fig. 15.2](#)).

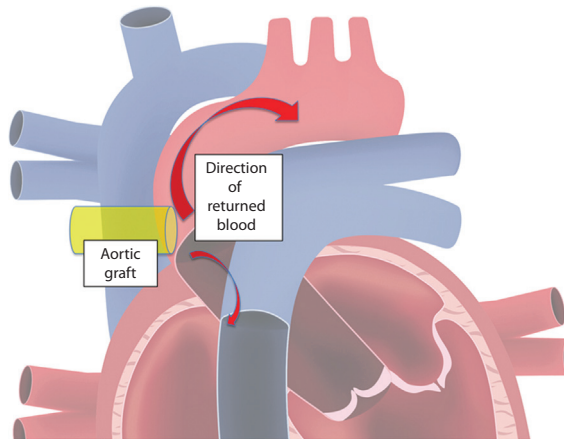


FIG. 15.2 Antegrade and retrograde blood flow from aortic graft.
(Modified from [Ody_Stocker/Shutterstock.com](#).)

Perfusion of the coronary arteries on the other hand is more dependent on retrograde flow from the cannula. While the majority of flow is directed in an antegrade direction, there is a small proportion of blood that flows back towards the heart which is needed for provision of oxygenated blood to the coronary arteries supplying the heart.

The second major advantage of this form of support is the amount of flow that can be provided. Remember that in peripherally inserted VA ECMO, we have a distinct limitation of flow that exists due to the retrograde nature of support. We can increase retrograde blood flow, but at a certain point, the advantages of higher ECMO blood flow (better perfusion/support) are outweighed by the disadvantages of retrograde support (left ventricular afterload, aortic valve closure, pulmonary edema).

The limits of retrograde flow to the overall amount of flow provided do not exist in central cannulation; therefore, central cannulation can be an option for a higher degree of support that can be considered on initiation of ECMO or as an upgrade of support. While retrograde limits to flow do not exist, limits to flow on central cannulation do exist.

LIMITS TO BLOOD FLOW ON CENTRAL CANNULATION

Let's envision a patient on central cannulation. As we review the clinical data, we note that the lactate is rising slightly, urine output is dropping off, mean arterial pressure is low, and venous oxygen saturation is low. We conclude that we are not supporting her optimally and want to increase our ECMO blood flow. What are the limits to flow in this patient?

Flow Limit #1: Limit of the ECMO Circuit/Pump

All of the considerations for flow limitation that we considered in [Chapter 10](#) still apply. There is still a maximal flow that the oxygenator can accommodate as well as the preload dependence/afterload sensitivity of the pump.

Preload dependence can still be a major limit to flow. Even though the cannula is placed directly into the right atrium and the collapsibility of the inferior vena cava (IVC) is not a limit to flow, a lack of venous return in the form of hypovolemia/bleeding can represent a major limit to flow.

Flow Limit #2: Mean Arterial Pressure

Afterload to the pump includes anything post pump to include tubing, oxygenator, cannulas, and ultimately mean arterial pressure. As flow increases, this can increase mean arterial pressure and increase the pressure differential that the pump will have to overcome. As this pressure differential increases, the pump may have to run at a higher rate of revolutions per minute (RPMs), which increases the risk of hemolysis.

Flow Limit #3: Pressures of the System

As flows across the pump increase, the post-pump pressure will increase as well. Higher pressures are sensed across the entire post-pump circuit, from the oxygenator all the way down to the cannula. The difference in pressure differential at normal (Fig. 15.3) and higher blood flows is illustrated in Fig. 15.4. The higher the blood flow, the higher the post-pump pressures will be (points 2 and 3).

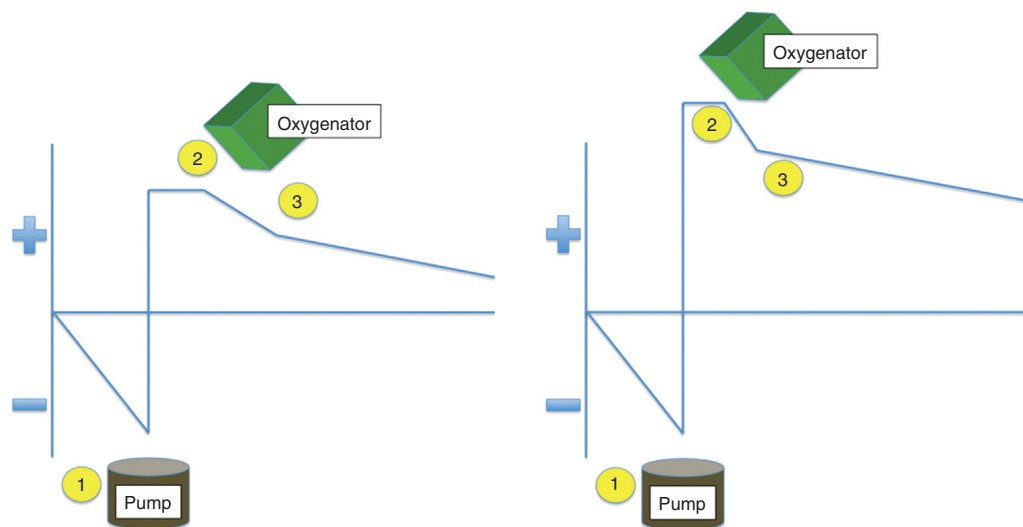


FIG. 15.3 and 15.4 Pressure differentials at normal blood flow and higher blood flow in central VA ECMO

The pressure limits of the system can be usually defined and set by individual facilities. The higher these pressures, the higher the risk of damage to the oxygenator, graft, or blood vessels into which the cannula is implanted.

Flow Limit #4: Overflow of the System

At a certain point, blood can overflow the arterial system, causing damage to blood vessels or end organs. In general, as with any type of support, excessive flow should be minimized. Rather, the focus should remain on maintaining the lowest possible blood flow that can achieve the desired physiologic endpoints.

LEFT VENTRICULAR VENTING IN CENTRAL CANNULATION

Just as with peripheral cannulation, venting of the left side of the heart should warrant careful consideration. Without venting, adverse effects of left ventricular distension can occur, to include pulmonary edema, left ventricular stasis, thrombosis, and increased wall stress/worsening of myocardial ischemia.

How Does Left Ventricular Distension Happen in Central Cannulation?

Left ventricular distension in central ECMO may not be immediately intuitive. The blood is ante-grade, and higher flows can allow for complete bypass of the pulmonary circulation. However, even if the blood flow of the ECMO circuit is high enough that there is no blood leaving the right ventricle into the pulmonary artery, there is blood that still drains into the heart. As we previously mentioned, the blood supply of the bronchial circulation comes off the aorta and drains directly into the pulmonary veins, which return blood to the left heart. In a similar way, some of the small cardiac endocardial veins, referred to as the Thebesian vein, drain blood directly into the left ventricle rather than into the coronary sinus.

It is these venous blood vessels that can contribute to ongoing left ventricular distension even if the heart is completely bypassed.

The left ventricle drain, whether it is from the pulmonary vein or the left ventricular apex, can be connected to the drainage side of the circuit, via Y connector. Even though it is “red” oxygenated blood, it is used as a drain as the negative pressure is needed to pull the blood. The drain should be connected to a clamp and flow probe, in order to regulate the amount of blood being drained. Enough blood should be drained from the left side of the heart to minimize the effects of left ventricular distension while simultaneously allowing for enough cardiac filling to allow for ejection. Regulating this will require careful examination for pulmonary edema as well as echocardiographic findings (to ensure sufficient draining) while also observing for pulsatility (to ensure sufficient filling).

REGULATION OF BLOOD FLOW ON CENTRAL VA ECMO

We discussed the limits to blood flow earlier. However, how much blood flow should be targeted; the maximum possible or some other target? The answer lies in the goals of support in the clinical situation.

The clinical spectrum may look something like the illustration in Fig. 15.5. Initially, the goal is just trying to maintain perfusion targets, then allowing for cardiac rest/buying time to wait for further takebacks/surgeries, and finally ending with cardiac recovery and ability to wean. Cardiac rest is the concept of giving the heart time to recover after its injury without worsening myocardial damage due to excessive filling or escalating doses of inotropes. The lower the blood flow, the more cardiac filling that we allow, with the aim of allowing for some contribution of the native heart and lungs.

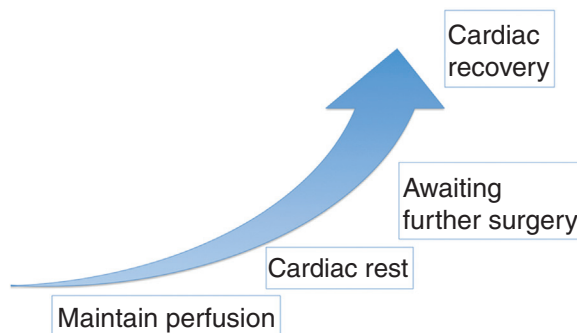


FIG. 15.5 Goals of central VA ECMO support

The other reason to allow for some cardiac activity is to allow for flow across the pulmonary vascular bed. As the ECMO blood flow increases and the amount of blood leaving the right heart

continues to decrease, there is progressive stasis of blood in the pulmonary arteries/veins. This will increase the risk of thrombosis even in the presence of anticoagulation.

REGULATION OF SWEEP GAS FLOW ON CENTRAL VA ECMO

The physiology of central antegrade flow requires more careful attention and regulation of the post-oxygenator blood. Normally, the post-oxygenator blood gas is used to evaluate the health of the *membrane* oxygenator, relying on the patient arterial blood gas to determine the effect of what is going on with the patient, after the mixing of native and ECMO circulation.

In central cannulation, the blood returning from the oxygenator will closely resemble what is leaving the oxygenator. This means that if the post-oxygenator PaO_2 is 350 mmHg, then we can likely anticipate that the blood gas results from the arterial line will also have a PaO_2 close to 350 mmHg, especially if ECMO blood flow is the primary contributor to overall circulation (Fig. 15.6).

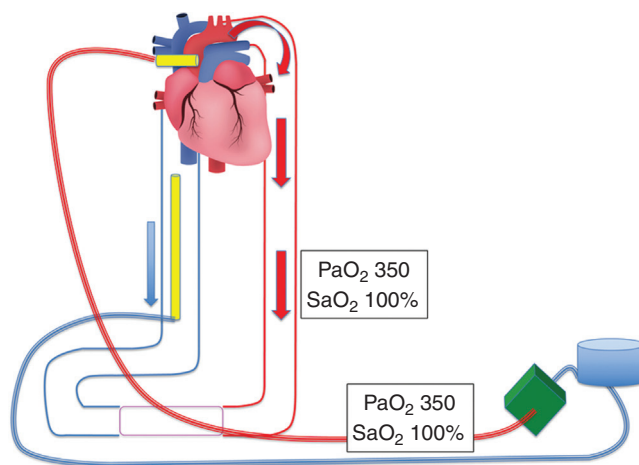


FIG. 15.6 Central cannulation and oxygen content of returned blood.
(Modified from [Ody_Stocker/Shutterstock.com](#).)

Why is this relevant? More oxygen is better, right?

Remember that optimal oxygen delivery requires well-saturated hemoglobin, with additional dissolved oxygen (PaO_2) contributing little to oxygen delivery. Excessive PaO_2 concentration, referred to as hyperoxia, does little to improve oxygen delivery to the peripheral tissue and can be harmful to the end organs, especially to the brain. This can be particularly relevant in the case of any brain that had any anoxic insult, such as in cardiac arrest, protracted hypoxia, or prolonged cardiopulmonary bypass run, all common in patients on central VA ECMO.

Also pay careful attention to CO_2 levels, adjusting sweep gas flow down as appropriate. In the same way that PaO_2 can easily be supraphysiologic in post-oxygenator blood, the efficiency of the membrane oxygenator in clearing CO_2 can lead to low levels of PaCO_2 in the post-oxygenator blood. Normally this does not matter as this blood mixes with CO_2 from the native circulation, giving a normalized CO_2 that can be adjusted with sweep gas flow. However, since there may be little to no mixing with native circulation in central VA ECMO, there is the possibility of hypocapnea in the arterial blood. Low arterial blood CO_2 levels can lead to further

brain injury, by causing the constriction of cerebral blood vessels and decreasing the cerebral perfusion.

Therefore, pay close attention to the post-oxygenator/patient arterial blood gases, aiming for normal levels of $\text{PaO}_2/\text{PaCO}_2$. Often, close regulation of oxygenation of the post-oxygenator blood needs to be made – this may be time to use the FiO_2 dial on the oxygen blender, dialing back the FiO_2 to aim for normal PaO_2 .

EFFECT OF NATIVE CARDIAC OUTPUT ON ANTEGRADE CENTRAL FLOW

The aforementioned phenomenon details the effects of central VA ECMO when the ECMO blood flow accounts for the majority of perfusion. However, as the heart recovers, there begins to be an effect of the native cardiac output. Let's explore how this manifests physiologically.

Let's say you are managing a patient on day 3 of his ECMO run. He was centrally cannulated for ECMO due to post-cardiotomy shock with a right atrial drain, an aortic return cannula, and a left ventricular apical vent. When you do your initial assessment, you note that the oxygen saturation is 88%.

HYPOXIA ON CENTRAL VA ECMO?

Although not as common as in peripheral VA ECMO, this can occur as the heart starts to recover and account for a greater percentage of overall perfusion. In this case, the blood flow from the heart/lungs is not competing against the blood flow from the ECMO circuit as with peripheral VA ECMO, but rather running in series with the ECMO circuit, like in VV ECMO (Fig. 15.7).

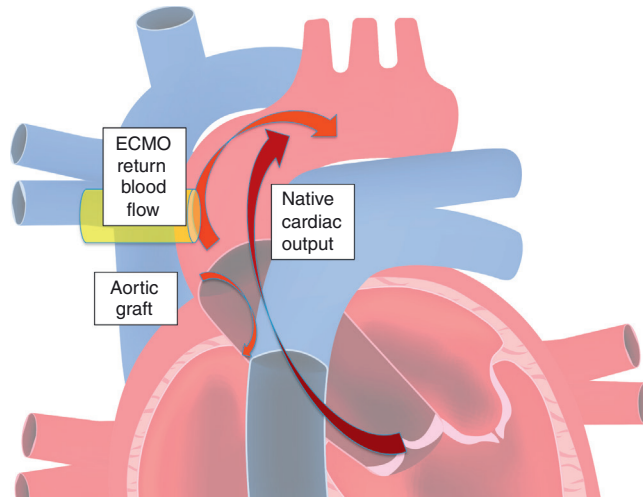


FIG. 15.7 Contribution of native cardiac output in a recovering heart on central VA ECMO. (Modified from [Ody_Stocker/Shutterstock.com](https://www.shutterstock.com/user/ody-stocker).)

The result is that much like VV ECMO, the ECMO circuit is only accounting for a proportion of the cardiac output. As the heart continues to recover, the saturation will more closely resemble the saturation of the blood returning to the left atrium from the pulmonary veins.

Although the oxygenation may not be optimal, this may represent a recovery of the cardiac function to such a degree that assessment for weaning may be possible.

PUTTING IT TOGETHER: THE RISKS/BENEFITS OF CENTRAL ECMO

To this point, central VA ECMO may seem like an attractive option. It is able to provide oxygenation, pulmonary, and cardiac support, without the strain on the right heart or left heart seen in VV ECMO or VA ECMO, respectively.

However, this does not mean that central ECMO is not without its disadvantages. The primary disadvantage involves the invasiveness of this configuration, often requiring sternotomy, access to heart/major blood vessels, and maintenance of an open chest. Complications can include bleeding, infection, stroke, need for subsequent operations, and heavy sedation/analgesia requirements. Additionally, maintaining an open chest can significantly limit mobility.

The benefits of antegrade flow should be weighed against these drawbacks/risks, when considering the configuration to support.

SUGGESTED READING

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- Raffa, G. M., Kowalewski, M., Brodie, D., Ogino, M., Whitman, G., & Meani, P., et al. (2019). Meta-analysis of peripheral or central extracorporeal membrane oxygenation in postcardiotomy and non-postcardiotomy shock. *The Annals of Thoracic Surgery*, 107(1), 311–321.